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Structural cabinet for an appliance incorporating unitary metallic boxes

Abstract

An appliance includes a metallic outer wrapper having sidewalls, a wrapper backwall and a machine wall. At least one metallic inner liner has liner walls and a liner backwall, wherein the metallic outer wrapper and the at least one metallic inner liner are coupled together at a trim breaker to define a structural cabinet having a hermetically sealed interior cavity defined between the metallic outer wrapper and the at least one metallic inner liner. The trim breaker defines a front face of the structural cabinet. At least one trim breaker conduit extends through the wrapper and liner backwalls wherein the trim breaker conduit defines a conduit through the structural cabinet, and wherein the structural cabinet is hermetically sealed at the trim breaker conduit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/297,867 filed Apr. 10, 2023, now U.S. Pat. No. 12,049,979, which is a continuation of U.S. patent application Ser. No. 17/553,010 filed Dec. 16, 2021, now U.S. Pat. No. 11,680,673, which is a continuation of U.S. patent application Ser. No. 17/064,853 filed Oct. 7, 2020, now U.S. Pat. No. 11,248,734, which is a continuation of U.S. patent application Ser. No. 16/309,320 filed Dec. 12, 2018, now U.S. Pat. No. 10,830,384, which is a national stage of International Application No. PCT/US2016/056389 filed Oct. 11, 2016, all of which are entitled STRUCTURAL CABINET FOR AN APPLIANCE INCORPORATING UNITARY METALLIC BOXES, now International Publication No. WO2018/070996, the entire disclosures of which are hereby incorporated herein by reference.

FIELD OF THE DEVICE

(1) The device is in the field of refrigerating appliances, and more specifically, a refrigerating appliance incorporating a welded metallic outer wrapper and welded metallic inner liners for forming a unitary structural cabinet.

SUMMARY

(2) In at least one aspect, an appliance includes a metallic outer wrapper having sidewalls, a wrapper backwall and a machine wall. At least one metallic inner liner has liner walls and a liner backwall, wherein the metallic outer wrapper and the at least one metallic inner liner are coupled together at a trim breaker to define a structural cabinet having a hermetically sealed interior cavity defined between the metallic outer wrapper and the at least one metallic inner liner. The trim breaker defines a front face of the structural cabinet. At least one trim breaker conduit extends through the wrapper and liner backwalls wherein the trim breaker conduit defines a conduit through the structural cabinet, and wherein the structural cabinet is hermetically sealed at the trim breaker conduit.

(3) In at least another aspect, an appliance includes a metallic outer wrapper having sidewalls, a wrapper backwall and a machine wall, at least one metallic inner liner having liner walls and a liner backwall, wherein the metallic outer wrapper and the at least one metallic inner liner are coupled together at a trim breaker to define a structural cabinet having a hermetically sealed interior cavity defined between the metallic outer wrapper and the at least one metallic inner liner. The trim breaker defines a front face of the structural cabinet. A cavity partition is disposed within the interior cavity, the cavity partition separating the interior cavity into separate and distinct first and second portions of the interior cavity.

(4) In at least another aspect, a method of forming an appliance cabinet includes welding sidewalls, a wrapper backwall and a machine wall together to define a metallic outer wrapper, wherein a wrapper conduit aperture is defined within the metallic outer wrapper. The method also includes welding liner sidewalls and a liner backwall together to define a metallic inner liner, wherein a liner conduit aperture is defined within the metallic inner liner. The method also includes disposing the metallic inner liner into the metallic outer wrapper and aligning the wrapper conduit aperture with the liner conduit aperture. The method also includes attaching a trim breaker to a liner edge of the inner liner and a wrapper edge of the outer wrapper to define a structural cabinet with an interior cavity defined between the inner liner and the outer wrapper. The method also includes attaching a trim breaker conduit to the wrapper and liner conduit apertures to define a conduit extending through the structural cabinet. The method also includes expressing gas from the interior cavity, wherein an at least partial vacuum is defined therein.

(5) In at least another aspect, a method of forming an appliance cabinet includes welding sidewalls, a wrapper backwall and a machine wall together to define a metallic outer wrapper. The method also includes welding liner sidewalls and a liner backwall together to define a metallic inner liner. The method also includes disposing the metallic inner liner into the metallic outer wrapper and disposing a cavity partition between the inner liner and the outer wrapper. The method also

includes attaching a trim breaker to a liner edge of the inner liner and a wrapper edge of the outer wrapper to define a structural cabinet with an interior cavity defined between the inner liner and the outer wrapper, wherein the cavity partition separates the interior cavity into first and second portions of the interior cavity. The method also includes expressing gas from the first portion of the interior cavity and expressing gas from the second portion of the interior cavity.

(6) These and other features, advantages, and objects of the present device will be further understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a front perspective view of a structural cabinet for an appliance showing door panels in an open position to view a refrigerating compartment;

(3) FIG. 2 is an exploded perspective view of an aspect of a structural cabinet for an appliance;

(4) FIG. 3 is an exploded perspective view of an outer wrapper for the structural cabinet of FIG. 2;

(5) FIG. 4 is an exploded perspective view of an inner liner for a refrigerating compartment of the structural cabinet of FIG. 2;

(6) FIG. 5 is an exploded perspective view of the inner liner for a freezing compartment of the structural cabinet of FIG. 2;

(7) FIG. 6 is a perspective view of a trim breaker conduit for forming a conduit through the structural cabinet of FIG. 2;

(8) FIG. 7 is a front elevational view of the trim breaker conduit of FIG. 6;

(9) FIG. 8 is a cross-sectional view of the trim breaker conduit of FIG. 7 taken along line VIII-VIII;

(10) FIG. 9 is a front elevational view of the structural cabinet of FIG. 2;

(11) FIG. 10 is a perspective cross-sectional view of the structural cabinet of FIG. 9, taken along line X-X;

(12) FIG. 11 is a perspective cross-sectional view of the structural cabinet of FIG. 9, taken along line XI-XI;

(13) FIG. 12 is a perspective cross-sectional view of the structural cabinet of FIG. 9, taken along line XII-XII;

(14) FIG. 13 is a perspective view of a structural cabinet incorporating a cavity partition for separating the interior cavity into distinct interior cavity portions;

(15) FIG. 14 is a cross-sectional view of a structural cabinet incorporating an aspect of the cavity partition;

(16) FIG. 15 is an exploded perspective view of the structural cabinet of FIG. 13;

(17) FIG. 16 is a linear flow diagram illustrating a method for forming an appliance cabinet; and

(18) FIG. 17 is a linear flow diagram illustrating a method for forming an appliance cabinet.

DETAILED DESCRIPTION OF EMBODIMENTS

(19) For purposes of description herein the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the device as oriented in FIG. 1.

However, it is to be understood that the device may assume various alternative orientations and step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(20) As illustrated in FIGS. 1-12, reference numeral 10 generally refers to a structural cabinet for

an appliance **12**. Such appliances **12** can include, but are not limited to, refrigerators, freezers, coolers, pantries, combinations thereof, ovens, laundry appliances, dishwashers, water heaters, and other similar appliances **12** found within residential and commercial settings. According to the various embodiments, the appliance **12** can include a metallic outer wrapper **14** having sidewalls **16**, a wrapper backwall **18**, and a machine wall **20**. At least one metallic inner liner **22** is also included. The at least one metallic inner liner **22** includes liner walls **24** and a liner backwall **26**. It is contemplated that the metallic outer wrapper **14** and the at least one metallic inner liner **22** can be coupled together at a trim breaker **28** to define the structural cabinet **10** having a hermetically sealed interior cavity **30** that is defined between the metallic outer wrapper **14** and the one or more metallic inner liners **22**. It is further contemplated that the trim breaker **28** can be positioned to define a front face **32** of the structural cabinet **10**. At least one trim breaker conduit **34** is positioned to extend through the outer wrapper **14** and inner liner **22**, such as the wrapper and liner backwalls **18**, **26**. In such an embodiment, the trim breaker conduit **34** defines a utility conduit **36** through the structural cabinet **10**. The trim breaker conduit **34** can be attached to the wrapper and liner backwalls **18**, **26** such that the structural cabinet **10** remains hermetically sealed at the trim breaker conduit **34** and at the trim breaker **28**.

(21) Referring again to FIGS. **1-12**, the trim breaker conduit **34** defines a sealed grommet-type opening that extends through the structural cabinet **10** to allow for various utilities **50** to be run through the structural cabinet **10** from a machine compartment **52** of the appliance **12**. It is contemplated that the machine compartment **52** can be disposed proximate the machine wall **20** of the metallic outer wrapper **14**. Wiring, piping, tubing, ductwork, and other utilities **50** are adapted to be run from the machine compartment **52** and through the trim breaker conduits **34** for providing paths within the utility conduits **36** through which these utilities **50** can be run within the appliance **12**. By incorporating the trim breaker conduits **34** within the structural cabinet **10**, openings, punctures, and other openings within the structural cabinet **10** can be kept to a minimum. Additionally, these utility conduits **36** within the structural cabinet **10** can be located at the trim breaker conduits **34**, such that an at least partial vacuum can be maintained within the interior cavity **30** of the structural cabinet **10**.

(22) According to the various embodiments, it is contemplated that an insulating material **60** can be disposed within the hermetically sealed interior cavity **30**. Additionally, the hermetically sealed interior cavity **30** can be placed under an at least partial vacuum to define a vacuum insulated structure of the structural cabinet **10**. By incorporating the various trim breaker conduits **34**, the at least partial vacuum can be maintained within the interior cavity **30**, while also allowing for the placement of utilities **50** through the utility conduits **36** within the appliance **12** and through the structural cabinet **10**.

(23) Referring again to FIGS. **1-12**, it is contemplated that each trim breaker conduit **34** can include a wrapper conduit channel **70** that receives a wrapper edge **72** proximate a wrapper aperture **74** of the outer wrapper **14**. Each trim breaker conduit **34** can also include a liner conduit channel **76** that receives a liner edge **78** proximate a liner aperture **80** within the inner liner **22**. The trim breaker conduit **34** can also include a conduit wall **82** that extends between the wrapper and liner conduit channels **70**, **76**. It is contemplated that the inner surface **84** of the conduit wall **82** defines the space of the utility conduit **36** through the structural cabinet **10** that allows for passage of the utilities **50** within the appliance **12**. The outer surface **86** of the conduit wall **82** serves to define a portion of the hermetically sealed interior cavity **30**.

(24) Referring again to FIGS. **1-12**, it is contemplated that the structural cabinet **10** can include first and second inner liners **100**, **102**. Each of the first and second inner liners **100**, **102** can include corresponding first and second trim breaker conduits **104**, **106**. It is contemplated that the first trim breaker conduit **104** defines a first utility conduit **108** through the structural cabinet **10** at the first inner liner **100**. The second trim breaker conduit **106** defines a second utility conduit **110** through the structural cabinet **10** at the second inner liner **102**. It is contemplated that the structural cabinet

10 can include the first and second trim breaker conduits **104**, **106** and can also include additional trim breaker conduits **34**. Typically, a single trim breaker conduit **34** will be disposed within the structural cabinet **10** at or proximate each interior compartment **112** that typically corresponds to a separate dedicated inner liner **22**.

(25) As exemplified in FIGS. 2-9, a first inner liner **100** corresponds to a refrigerating compartment **120** and a second inner liner **102** corresponds to a freezing compartment **122**. Typically, the conduit wall **82** for each of the trim breaker conduits **34** is adapted to extend perpendicularly between the wrapper and liner conduit channels **70**, **76** of each trim breaker conduit **34**. In this manner, the conduit is adapted to define a grommet that extends perpendicularly through the structural cabinet **10**. It is also contemplated that various aspects of the trim breaker conduit **34** can extend at an angle through the structural cabinet **10** where such angle is useful to define a more convenient path through the structural cabinet **10** for delivering various utilities **50** to certain areas of the appliance **12**.

(26) Referring again to FIGS. 1-12, it is contemplated that the sidewalls **16**, the wrapper backwall **18** and the machine wall **20** of the outer wrapper **14** are each metallic members that are welded together to define the metallic outer wrapper **14** as a single unitary piece. Similarly, the liner walls **24** and liner backwall **26** of each of the inner liners **22** can also be metallic members that are welded together. In this manner, each inner liner **22** defines a metallic inner liner **22** as a single unitary piece. Accordingly, the outer wrapper **14** defines an outside box that is formed through the welded engagement of three metallic components: the sidewalls **16**, the wrapper backwall **18** and the machine wall **20**.

(27) Referring again to FIGS. 2, 3 and 9, it is contemplated that the sidewall **16** of the outer wrapper **14** is a metallic part that can be bent into a rectangular shape and welded at side joints **130**. The wrapper backwall **18** is a metallic sheet that is formed with a backwall edge **132** that engages the sidewalls **16**. The wrapper backwall **18** can include a machine aperture **134** through which the machine wall **20** can be placed and welded to the wrapper backwall **18**. It is contemplated that the wrapper apertures **74** that receive the wrapper conduit channel **70** can be disposed in various portions of the outer wrapper **14**. Typically, the wrapper conduit aperture will be disposed within the machine wall **20** and/or the wrapper backwall **18**.

(28) Referring again to FIGS. 2, 4, 5 and 9, the metallic inner liners **22** are formed in a similar fashion to that of the outer wrapper **14**. Each of the inner liners **22** can include the liner walls **24** that form a substantially perimetrical shape that is welded to the liner backwall **26**. It is contemplated that the liner backwall **26** can include a planar member or a member that is contoured to match the general shape of certain portions of the appliance **12**. By way of example, and not limitation, the liner backwall **26** of the second inner liner **102** can be shaped to match the contour of the machine wall **20** of the outer wrapper **14**. In this manner, the machine compartment **52** can be positioned behind the freezing compartment **122** of the structural cabinet **10** while the second inner liner **102** and outer wrapper **14** can also define a portion of the interior cavity **30** that extends around the machine compartment **52** and also provides insulating functions around the machine compartment **52**.

(29) Referring again to FIGS. 1-12, the outer wrapper **14** and inner liners **22** are attached together by placing the various inner liners **22** within the outer wrapper **14**. The trim breaker **28** includes a wrapper channel **140** that engages a wrapper edge **72** of the outer wrapper **14**. The trim breaker **28** also includes a liner channel **142** that receives a liner edge **78** of each of the inner liners **22**. It is contemplated that the trim breaker **28** can attach to the outer wrapper **14** and inner liners **22** through an adhesive, or can be attached through various welding and/or mechanical attachment processes, or combinations of adhesive, welding and/or mechanical attachment.

(30) Referring again to FIGS. 1-12, it is contemplated that where the first and second inner liners **100**, **102** come together, an interior mullion **150** extends between the first and second inner liners **100**, **102**. A portion of the trim breaker **28** can be integral to extend across the interior mullion **150**.

In this manner, the trim breaker **28** defines the front face **32** of the structural cabinet **10**. The trim breaker **28** can also include various utility features proximate the front face **32** such as a heat loop **152**, structural attachment points for hinges **154**, and an interior insulating space **156** that receives at least a portion of the insulating material **60**. It is contemplated that the trim breaker **28** can at least partially define the hermetically sealed interior cavity **30** of the structural cabinet **10**. To achieve the at least partial vacuum within the interior cavity **30**, it is contemplated that a hermetic seal can be defined between the trim breaker **28**, the outer wrapper **14** and the various inner liners **22** of the appliance **12**. To further define this hermetic seal, it is contemplated that the trim breaker **28** and the trim breaker conduits **34** of the structural cabinet **10** can be made of metal, plastic, various polymers, composite materials, combinations thereof, and other similar materials that can be formed to create a hermetic seal within the interior cavity **30** of the structural cabinet **10**.

(31) Referring now to FIGS. **1-15**, it is contemplated that the appliance **12** can include the metallic outer wrapper **14** having sidewalls **16**, the wrapper backwall **18** and the machine wall **20** that are welded together to define the unitary metallic outer wrapper **14**. One or more metallic inner liners **22** can also be included, where each metallic inner liner **22** includes liner walls **24** and a liner backwall **26** that are welded together to define unitary metallic inner liners **22** that are disposed within the metallic outer wrapper **14**. It is contemplated that the metallic outer wrapper **14** and the various metallic inner liners **22** are coupled together to define the structural cabinet **10** that includes the hermetically sealed interior cavity **30** defined between the metallic outer wrapper **14** and the various metallic inner liners **22**. As discussed above, the trim breaker **28** typically defines a front face **32** of the structural cabinet **10**. A cavity partition **170** is disposed within the interior cavity **30**. The cavity partition **170** is adapted to separate the interior cavity **30** into separate and distinct first and second portions **172**, **174** of the interior cavity **30**. It is contemplated that the cavity partition **170** maintains the first portion **172** of the interior cavity **30** hermetically sealed and separate from the second portion **174** of the interior cavity **30**.

(32) According to the various embodiments, by separating the first and second portions **172**, **174** of the interior cavity **30** through use of the cavity partition **170**, each of the first and second portions **172**, **174** can be maintained as a separate vacuum insulated structure. Accordingly, damage to one of the first and second portions **172**, **174** that may result in loss of vacuum within that portion will not affect the at least partial vacuum within the other of the first and second portions **172**, **174** of the interior cavity **30**. During formation of the structural cabinet **10**, it is contemplated that the first portion **172** of the interior cavity **30** can be placed in communication with a first gas outlet **180** for expressing gas **182** from the first portion **172**. Similarly, the second portion **174** of the interior cavity **30** is placed in communication with a second gas outlet **184** for separately expressing gas **182** from the second portion **174** of the interior cavity **30**. In this manner, different air pressures can be defined within each of the first and second portions **172**, **174** of the interior cavity **30**.

(33) Referring again to FIGS. **1-15**, where the structural cabinet **10** includes first and second inner liners **100**, **102**, it is contemplated that the cavity partition **170** can extend between the first and second inner liners **100**, **102**. In this manner, the first inner liner **100** can partially define the first portion **172** of the interior cavity **30**, but not the second portion **174**. Conversely, the second inner liner **102** can partially define the second portion **174** of the interior cavity **30**, but not the first portion **172**. In such an embodiment, the cavity partition **170** can extend from one sidewall **16** of the outer wrapper **14** to the other sidewall **16** of the outer wrapper **14** and defines surfaces of each of the first and second portions **172**, **174**. The cavity partition **170** extends between the inner liner **22** and the outer wrapper **14** and engages a portion of the trim breaker **28** proximate the interior mullion **150**. Accordingly, the cavity partition **170** and the interior mullion **150** serves to define a boundary and/or dividing point between the first and second portions **172**, **174** of the interior cavity **30**. It is also contemplated that the cavity partition **170** can be contained within a wall **190** of the structural cabinet **10** and can extend between an inner liner **22** of the structural cabinet **10** and the outer wrapper **14** of the structural cabinet **10**. In such an embodiment, a cavity partition **170** is a

generally U-shaped member that extends through the wall **190** of the structural cabinet **10**. It is contemplated that a plurality of cavity partitions **170** can be included within the structural cabinet **10**. In this manner, the interior cavity **30** can be separated into a plurality of interior cavity portions **192** that are each hermetically sealed from one another. Accordingly, loss of vacuum within any one of the interior cavity portions **192** of the insulating cavity may not result in a loss of vacuum within other interior cavity portions **192** of the interior cavity **30**.

(34) Through the use of the one or more cavity partitions **170** within the structural cabinet **10**, it is contemplated that service calls can be made with respect to one or more interior cavity portions **192** of the insulating cavity. While waiting for such service call, the insulating capabilities of the other interior cavity portions **192** of the insulating cavity may be unaffected as the at least partial vacuum within these other portions can be maintained even though one of the other interior portions of the interior cavity **30** may be damaged. It is contemplated that pressure sensors can be disposed within the various interior cavity portions **192** of the interior cavity **30** to monitor the level of vacuum pressure within each interior cavity portion **192** of the interior cavity **30**. A loss of pressure in any one interior cavity portion **192** can be communicated to the user or to a service center to alert that a service call may be needed.

(35) According to the various embodiments, the cavity partition **170** can be made of any one of various materials that can include, but is not limited to, metals, plastics, polymers, composite materials, combinations thereof, and other similar materials that can maintain the various interior cavity portions **192** of the interior cavity **30** hermetically sealed and separate from one another.

(36) According to the various embodiments, it is contemplated that the interior mullion **150** can include a partition channel **198** proximate the wrapper channel **140** and liner channel **142** that receives a partition edge **200** of the cavity partition **170**. In this manner, the cavity partition **170** can also engage a portion of the trim breaker **28** to define the hermetically separated first and second portions **172**, **174** of the interior cavity **30** of the structural cabinet **10**.

(37) Referring now to FIGS. **1-16**, having described various aspects of the structural cabinet **10**, a method **400** is disclosed for forming a structural cabinet **10** for an appliance **12**. According to the method **400**, the outer wrapper **14** is formed by welding the sidewalls **16**, the wrapper backwall **18** and the machine wall **20** together to define the metallic outer wrapper **14** (step **402**). At least one wrapper aperture **74** can be defined within the metallic outer wrapper **14**. The liner sidewalls **16** and liner backwall **26** can also be welded together to define one or more metallic inner liners **22** (step **404**). A liner conduit aperture can also be defined within at least one of the metallic inner liners **22**. Typically, each metallic inner liner **22** will include a dedicated liner aperture **80** for receiving a trim breaker conduit **34** to define a conduit through the structural cabinet **10**. The metallic inner liner **22** is then disposed into the metallic outer wrapper **14** (step **406**). The inner liner **22** and outer wrapper **14** are aligned, such that the wrapper aperture **74** is disposed in alignment with the liner aperture **80** for proper disposition of the trim breaker conduit **34** within the wrapper and liner apertures **74**, **80**. The trim breaker **28** is then attached to the liner edge **78** of the inner liner **22** and the wrapper edge **72** of the outer wrapper **14** to define the structural cabinet **10** with the interior cavity **30** defined therebetween (step **408**). The trim breaker conduit **34** is also attached to the wrapper and liner apertures **74**, **80** to define a utility conduit **36** extending through the structural cabinet **10** (step **410**). As discussed above, the trim breaker **28** and the trim breaker conduit **34** can be hermetically sealed to the inner liner **22** and outer wrapper **14** to define the hermetically sealed interior cavity **30**. Once the hermetic seal is formed, gas **182** can be expressed, expelled, or otherwise removed from the interior cavity **30** to define a vacuum insulated structure of the structural cabinet **10** (step **412**). In this manner, an at least partial vacuum is defined within the interior cavity **30**.

(38) Referring again to FIGS. **1-16**, it is contemplated that the step **406** of disposing the metallic inner liner **22** into the metallic outer wrapper **14** can also include disposing the cavity partition **170** between the inner liner **22** and the outer wrapper **14**. As discussed above, the cavity partition **170**

can be adapted to separate the interior cavity **30** into first and second portions **172**, **174** of the interior cavity **30**. These first and second portions **172**, **174** of the interior cavity **30** can be maintained hermetically separate from one another such that separate vacuums and potentially different vacuum pressures can be defined within each of the first and second portions **172**, **174**, and additional portions, of the interior cavity **30**. It is further contemplated that the step **412** of expressing, expelling or otherwise removing gas **182** from the interior cavity **30** can include expressing gas **182** from the first portion **172** of the interior cavity **30** via the first gas outlet **180** and separately removing gas **182** from the second portion **174** of the interior cavity **30** via the second gas outlet **184**.

(39) Referring now to FIGS. **1-15** and **17**, a method **500** is disclosed for forming a structural cabinet **10** for an appliance **12**. According to the method **500**, sidewalls **16**, a wrapper backwall **18** and a machine wall **20** are welded together to define the metallic outer wrapper **14** (step **502**). Liner sidewalls **16** and a liner backwall **26** are also welded together to define a metallic inner liner **22** (step **504**). As discussed above, one or more inner liners **22** can be formed, depending on the number of interior compartments **112** that are included within the structural cabinet **10**. According to the method **500**, the metallic inner liner **22** is disposed into the metallic outer wrapper **14** (step **506**). The cavity partition **170** can be disposed between the inner liner **22** and the outer wrapper **14** (step **508**). It is also contemplated that the cavity partition **170** can be disposed between separate inner liners **22**, such as through an interior mullion **150** of the structural cabinet **10**. According to the method **500**, the trim breaker **28** is attached to a liner edge **78** of the inner liner **22** and a wrapper edge **72** of the outer wrapper **14** to define the structural cabinet **10** with an interior cavity **30** defined therebetween (step **510**). Through forming the structural cabinet **10**, the cavity partition **170** separates the interior cavity **30** into first and second portions **172**, **174** of the interior cavity **30**. After forming the structural cabinet **10**, gas **182** can be expressed from the first portion **172** of the interior cavity **30** (step **512**) and gas **182** can also be separately expressed from the second portion **174** of the interior cavity **30** (step **514**).

(40) Referring again to FIGS. **1-17**, in each of the embodiments of the structural cabinet **10** that may or may not include an aspect of the cavity partition **170**, the wrapper and liner apertures **74**, **80**, such as conduit apertures, can be formed within the outer wrapper **14** and one or more inner liner **22**, respectively. The use of the trim breaker conduit **34** attached to the wrapper and liner apertures **74**, **80** defines the utility conduit **36** extending through the structural cabinet **10** for allowing various utilities **50** to pass through the structural cabinet **10** without puncturing or otherwise damaging the at least partial vacuum within the interior cavities **30** and hermetically separate portions of the interior cavity **30**.

(41) It will be understood by one having ordinary skill in the art that construction of the described device and other components is not limited to any specific material. Other exemplary embodiments of the device disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(42) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(43) It is also important to note that the construction and arrangement of the elements of the device as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of

parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(44) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present device. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(45) It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present device, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

(46) The above description is considered that of the illustrated embodiments only. Modifications of the device will occur to those skilled in the art and to those who make or use the device. Therefore, it is understood that the embodiments shown in the drawings and described above is merely for illustrative purposes and not intended to limit the scope of the device, which is defined by the following claims as interpreted according to the principles of patent law, including the Doctrine of Equivalents.

Claims

1. A method of forming an appliance cabinet, the method comprising steps of: welding sidewalls together to define a metallic outer wrapper having a wrapper conduit aperture; welding liner walls together to define a metallic inner liner having a liner conduit aperture; aligning the wrapper conduit aperture with the liner conduit aperture; attaching a trim breaker to a liner edge of the metallic inner liner and a wrapper edge of the metallic outer wrapper to define a structural cabinet; and attaching a trim breaker conduit to a wrapper edge at the wrapper conduit aperture and a liner edge at the liner conduit aperture to define a conduit extending through the structural cabinet, wherein the wrapper edge at the wrapper conduit aperture is perpendicular to the sidewall having the wrapper conduit aperture and wherein the liner edge at the liner conduit aperture is parallel with the wrapper edge at the wrapper conduit aperture.
2. The method of claim 1, wherein the step of attaching the trim breaker includes forming an interior cavity defined between the metallic inner liner, the metallic outer wrapper, the trim breaker, and the trim breaker conduit.
3. The method of claim 1, wherein the step of attaching the trim breaker conduit includes: bending the liner edge at the liner conduit aperture; and bending the wrapper edge at the wrapper conduit aperture.
4. The method of claim 2, further comprising the step of: expressing gas from the interior cavity, wherein an at least partial vacuum is defined therein.
5. The method of claim 4, wherein the step of expressing gas from the interior cavity includes expressing gas from a first portion of the interior cavity via a first gas outlet and separately

expressing gas from a second portion of the interior cavity via a second gas outlet.

6. The method of claim 1, wherein the step of welding the liner walls together to form the metallic inner liner includes forming a first inner liner and a second inner liner that cooperatively form the metallic inner liner.

7. The method of claim 6, wherein the step of aligning the wrapper conduit aperture with the liner conduit aperture includes disposing a cavity partition within an interior cavity defined between the first inner liner, the second inner liner, and the metallic outer wrapper.

8. The method of claim 7, wherein the cavity partition is disposed between the first inner liner and the second inner liner.

9. The method of claim 7, wherein the cavity partition separates the interior cavity into first and second portions of the interior cavity.

10. The method of claim 7, wherein the cavity partition is configured to attach to a portion of the trim breaker.

11. A method of forming an appliance cabinet, the method comprising steps of: welding sidewalls together to define a metallic outer wrapper, wherein a wrapper conduit aperture is defined within the metallic outer wrapper; welding liner walls together to define a metallic inner liner, wherein a liner conduit aperture is defined within the metallic inner liner; attaching a trim breaker to a liner edge of the metallic inner liner and a wrapper edge of the metallic outer wrapper to define a structural cabinet with an interior cavity defined between the metallic inner liner and the metallic outer wrapper, wherein the wrapper conduit aperture and the liner conduit aperture are aligned with one another; bending the liner edge at the liner conduit aperture; bending the wrapper edge at the wrapper conduit aperture, wherein the wrapper edge at the wrapper conduit aperture is perpendicular to the sidewall having the wrapper conduit aperture; and attaching a trim breaker conduit to the wrapper edge at the wrapper conduit aperture and the liner edge at the liner conduit aperture to define a conduit extending through the structural cabinet, wherein the liner edge at the liner conduit aperture is parallel with the wrapper edge at the wrapper conduit aperture.

12. The method of claim 11, wherein the step of welding the liner walls together to form the metallic inner liner includes forming a first inner liner and a second inner liner that cooperatively form the metallic inner liner.

13. The method of claim 12, wherein the step of attaching the trim breaker to the liner edge of the metallic inner liner and the wrapper edge of the metallic outer wrapper includes disposing a cavity partition between the metallic inner liner and the metallic outer wrapper.

14. The method of claim 13, wherein the cavity partition is disposed between the first inner liner and the second inner liner.

15. The method of claim 13, wherein the cavity partition separates the interior cavity into first and second portions of the interior cavity.

16. The method of claim 13, wherein the cavity partition is configured to attach to the trim breaker.

17. The method of claim 15, further comprising the step of: expressing gas from the interior cavity, wherein an at least partial vacuum is defined therein.

18. The method of claim 17, wherein the step of expressing gas from the interior cavity includes expressing gas from the first portion of the interior cavity via a first gas outlet and separately expressing gas from the second portion via a second gas outlet.

19. A method of forming a structural cabinet, the method comprising steps of: welding sidewalls together to define a metallic wrapper, wherein a wrapper conduit aperture is defined within the metallic wrapper; welding liner walls together to define a metallic liner, wherein a liner conduit aperture is defined within the metallic liner; bending a liner edge at the liner conduit aperture; bending a wrapper edge at the wrapper conduit aperture, wherein the wrapper edge at the wrapper conduit aperture is perpendicular to the sidewall having the wrapper conduit aperture; attaching a trim breaker to a liner edge of the metallic liner and a wrapper edge of the metallic wrapper, wherein the wrapper conduit aperture is positioned in alignment with the liner conduit aperture; and

attaching a trim breaker conduit to the wrapper edge at the wrapper conduit aperture and the liner edge at the liner conduit aperture to define a conduit extending through the structural cabinet to define an interior cavity defined between the metallic liner, the metallic wrapper, the trim breaker and the trim breaker conduit, wherein the liner edge at the liner conduit aperture is parallel with the wrapper edge at the wrapper conduit aperture.

20. The method of claim 19, wherein the step of attaching the trim breaker to the liner edge of the metallic liner and the wrapper edge of the metallic wrapper includes disposing a cavity partition between the metallic liner and the metallic wrapper, wherein the cavity partition separates the interior cavity into first and second portions of the interior cavity.
